$\boxed{P4.18}$ let $\begin{cases} L_+ f_\ell^m = A_\ell^m f_\ell^{m+1} \\ L_- f_\ell^m = B_\ell^m f_\ell^{m-1} \end{cases}$ find A_ℓ^m, B_ℓ^m.

Use the fact that $\begin{cases} L^2 f_\ell^m = \hbar \ell(\ell+1) f_\ell^m \\ L_z f_\ell^m = \hbar m f_\ell^m \end{cases}$

and $L^2 = L_\pm L_\mp + L_z^2 \mp \hbar L_z$

To find A_ℓ^m form: $L^2 f_\ell^m = (L_- L_+ + L_z^2 + \hbar L_z) f_\ell^m$

$$\Rightarrow \langle f_\ell^m | L^2 f_\ell^m \rangle = \langle f_\ell^m | L_- L_+ f_\ell^m \rangle +$$

$$+ \langle f_\ell^m | L_z^2 f_\ell^m \rangle + \hbar \langle f_\ell^m | L_z f_\ell^m \rangle =$$

$$= \langle L_-^\dagger f_\ell^m | L_+ f_\ell^m \rangle + (\hbar m)^2 \langle f_\ell^m | f_\ell^m \rangle + \hbar \cdot \hbar m \langle f_\ell^m | f_\ell^m \rangle$$

But $L_-^\dagger = (L_x - iL_y)^\dagger = L_x + iL_y = L_+$ because $L_x^\dagger = L_x, L_y^\dagger = L_y$

$$\Rightarrow \hbar^2 \ell(\ell+1) = \underbrace{\langle L_+ f_\ell^m | L_+ f_\ell^m \rangle}_{\langle A_\ell^m f_\ell^{m+1} | A_\ell^m f_\ell^{m+1} \rangle = |A_\ell^m|^2} + \hbar^2 m^2 + \hbar^2 m$$

$$\Rightarrow |A_\ell^m|^2 = \hbar^2 (\ell(\ell+1) - m(m+1))$$

$$A_\ell^m = \hbar \sqrt{\ell(\ell+1) - m(m+1)}$$

Thus for $m = \ell$: $A_\ell^\ell = \hbar \sqrt{\ell(\ell+1) - \ell(\ell+1)} = 0$ so that $L_+ f_\ell^\ell = 0$ as it should be.

Similarly, taking $L^2 = L_+ L_- + L_z^2 - \hbar L_z$ we find that

$$|B_\ell^m|^2 = \hbar^2 (\ell(\ell+1) - m(m-1)) \quad ; \quad B_\ell^m = \hbar \sqrt{\ell(\ell+1) - m(m-1)}$$

so that for $m = -\ell$: $L_- f_\ell^{-\ell} = \hbar \sqrt{\ell(\ell+1) - (-\ell)(-\ell-1)} f_\ell^{-\ell-1} = 0.$

P4.22 (a) $L_+ Y_\ell^\ell = \hbar\sqrt{\ell(\ell+1) - \ell(\ell+1)}\; Y_\ell^{\ell+1} = 0.$

(b) Eq. 4.130 : $L_+ = \hbar e^{+i\varphi}\left(\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\varphi}\right)$

$$L_z = \frac{\hbar}{i}\frac{\partial}{\partial\varphi}$$

Since $L_z Y_\ell^\ell(\theta,\varphi) = \hbar\ell Y_\ell^\ell(\theta,\varphi)$

$$\Rightarrow \frac{\hbar}{i}\frac{\partial}{\partial\varphi}Y_\ell^\ell(\theta,\varphi) = \hbar\ell\, Y_\ell^\ell(\theta,\varphi)$$

$$\Rightarrow Y_\ell^\ell(\theta,\varphi) = F(\theta)\, e^{i\ell\varphi}$$

Then $L_+ Y_\ell^\ell(\theta,\varphi) = \hbar e^{i\varphi}\left(\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\varphi}\right) F(\theta)\, e^{i\ell\varphi} = 0$

$$\Rightarrow e^{i\varphi}\, e^{i\ell\varphi}\left[\frac{\partial}{\partial\theta} + i\cot\theta\,(i\ell)\right] F(\theta) = 0$$

$$\Rightarrow \frac{dF}{d\theta} = \ell\cot\theta\, F(\theta) = \ell\frac{\cos\theta}{\sin\theta}\, F(\theta)$$

$$\Rightarrow \frac{dF}{F} = \ell\frac{\cos\theta}{\sin\theta}\, d\theta = \ell\frac{d(\sin\theta)}{\sin\theta} = d\big(\ln(\sin\theta)\big)\cdot\ell$$

$$\Rightarrow \ln F(\theta) = \ell\ln(\sin\theta) = \ln(\sin\theta)^\ell$$

$$\Rightarrow F(\theta) = (\sin\theta)^\ell \cdot C_\ell$$

normalization constant

Thus $\boxed{Y_\ell^\ell(\theta,\varphi) = C_\ell\,(\sin\theta)^\ell\, e^{i\ell\varphi}}$

(c) Require $1 = \int |Y_\ell^\ell|^2 \sin\theta\, d\theta\, d\varphi \Rightarrow$

$$\Rightarrow 1 = |C_\ell|^2 \int_0^\pi (\sin\theta)^{2\ell}\sin\theta\, d\ell \int_0^{2\pi} d\varphi = 2\pi|C_\ell|^2 \cdot$$

$$\cdot \int_{-1}^{1} (\sin^2\theta)^\ell\, d(-\cos\theta)$$

Introduce $x = -\cos\theta$, $dx = d(-\cos\theta) = \sin\theta\, d\theta$

$$\sin^2\theta = 1 - \cos^2\theta = 1 - x^2$$

$$\Rightarrow \int_{-1}^{1} (\sin^2\theta)^\ell \, d(-\cos\theta) = \int_{-1}^{1} (1-x^2)^\ell \, dx$$

$$\Rightarrow 1 = 2\pi |C_\ell|^2 \cdot \underbrace{2\int_0^1 (1-x^2)^\ell \, dx}_{I(\ell)}$$

let $y = x^2 \Rightarrow dy = 2x\,dx \Rightarrow dx = \dfrac{dy}{2\sqrt{y}}$

$$\Rightarrow I(\ell) = 2 \cdot \int_0^1 (1-x^2)^\ell \, dx = \int_0^1 (1-y)^\ell \, y^{-1/2} \, dy =$$

$$= \int_0^1 (1-y)^{\ell+1-1} \, y^{1/2-1} \, dy = B\left(\ell+1, \tfrac{1}{2}\right)$$

where beta-function $B(a,b) \equiv \dfrac{\Gamma(a)\,\Gamma(b)}{\Gamma(a+b)}$ and Γ = gamma function

Thus $I(\ell) = \dfrac{\Gamma(\ell+1)\,\Gamma(\frac{1}{2})}{\Gamma(\ell+1+\frac{1}{2})} = \dfrac{\Gamma(\frac{1}{2})\,\Gamma(1+\ell)}{\Gamma(\frac{3}{2}+\ell)}$

Now, $\Gamma(n) = (n-1)!$; $\cancel{\Gamma(\tfrac{1}{2}+n) = \dfrac{\Gamma(2n)}{\Gamma(n)} \dfrac{\sqrt{2\pi}}{2^{2n-1/2}}}$, $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

$\Gamma(\frac{1}{2}+n) = \sqrt{\pi}\,\dfrac{(2n-1)!!}{2^n}$

$$\Rightarrow I(\ell) = \frac{\sqrt{\pi}\;\ell!}{\sqrt{\pi}\;\dfrac{(2(1+\ell)-1)!!}{2^{1+\ell}}} = \frac{2^{1+\ell}\,\ell!}{(2\ell+1)!!}$$

$\Gamma(\frac{3}{2}+\ell) = \Gamma(\frac{1}{2}+1+\ell)$

But double factorial $(2\ell+1)!! = \dfrac{(2\ell+1)!}{2^\ell \ell!}$

$$\Rightarrow I(\ell) = \frac{2^{2\ell+1}(\ell!)^2}{(2\ell+1)!} \Rightarrow C_\ell = \frac{1}{\sqrt{2\pi}}\sqrt{\frac{(2\ell+1)!}{2^{2\ell+1}(\ell!)^2}}$$ agrees with Table 4.3

$\ell=0 \Rightarrow C_\ell = \left(\dfrac{1}{4\pi}\right)^{1/2}$; $\ell=1 \Rightarrow C_\ell = \left(\dfrac{3!}{4\pi\cdot 2^2}\right)^{1/2} = \left(\dfrac{3}{8\pi}\right)^{1/2}$; $\ell=2 \Rightarrow C_\ell = \sqrt{\dfrac{5!}{4\pi\cdot 2^4 (2!)^2}} = \sqrt{\dfrac{15}{32\pi}}$; $\ell=3 \Rightarrow C_\ell = \sqrt{\dfrac{7!}{4\pi\cdot 2^6 (2\cdot 3)^2}} = \sqrt{\dfrac{35}{64\pi}}$

P 4.44 (a) $n=4,\ \ell=3,\ m=3$

$$\psi_{433}(r,\theta,\varphi) = R_{43}(r)\, Y_3^3(\theta,\varphi)$$

From (4.32) $$Y_3^3 = (-1)^3\sqrt{\frac{2\cdot 3+1}{4\pi}\,\frac{(3-3)!}{(3+3)!}}\; e^{i3\varphi}\, P_3^3(\cos\theta)$$

$$P_3^3(\cos\theta) = 15\sin\theta\,(1-\cos^2\theta) = 15\sin^3\theta \quad \text{(Table 4.2)}$$

$$\Rightarrow Y_3^3 = (-1)\sqrt{\frac{7}{4\pi}\,\frac{1}{1\cdot2\cdot3\cdot4\cdot5\cdot6}}\; 15\sin^3\theta\, e^{i3\varphi}$$

$$6! = 2\cdot3\cdot(2\cdot2)\cdot5\cdot(3\cdot2) = 2^4\,3^2\,5$$

$$Y_3^3 = (-1)\sqrt{\frac{7}{4\pi}\,\frac{1}{5}}\;\frac{15}{2^2\cdot3}\sin^3\theta\, e^{i3\varphi} = -\frac{1}{4}\sqrt{\frac{35}{4\pi}}\sin^3\theta\, e^{i3\varphi}$$

$$= -\sqrt{\frac{35}{64\pi}}\sin^3\theta\, e^{+i3\varphi} \quad \text{(Table 4.3)}$$

While $$R_{43}(r) = \frac{1}{768\sqrt{35}}\, a^{-3/2}\left(\frac{r}{a}\right)^3 e^{-r/4a} \quad \text{(Table 4.7)}$$

$$\Rightarrow \psi_{433} = -\frac{1}{768\, a^{3/2}}\,\frac{1}{\sqrt{64\pi}}\left(\frac{r}{a}\right)^3 e^{-r/4a}\sin^3\theta\, e^{i3\varphi}.$$

(b) $$\langle r\rangle = \int r\,|\psi_{433}|^2\, r^2\,dr\,\sin\theta\,d\theta\,d\varphi =$$

$$= \int r^3|R_{43}|^2\,dr\;\underbrace{\int |Y_3^3|^2\sin\theta\,d\theta\,d\varphi}_{1}$$

$$= \frac{1}{(768\sqrt{35})^2}\,\frac{1}{a^3}\int_0^\infty r^3\left(\frac{r}{a}\right)^6 e^{-r/2a}\,dr =$$

$$= \left(x=\frac{r}{a}\right) = \frac{a}{(768)^2\cdot35}\int_0^\infty x^9 e^{-\frac{x}{2}}\,dx$$

Use $\int_0^\infty x^n e^{-x/a}\,dx = n!\,a^{n+1}$

$$\Rightarrow \langle r \rangle = \frac{a}{35\,(768)^2}\; 9!\; 2^{9+1} = \frac{9!\,2^{10}}{35\,(768)^2} \cdot a = \boxed{18a}$$

↳ Bohr radius

(c) $L_x^2 + L_y^2 = L^2 - L_z^2$

but $\psi_{n\ell m}$ is an eigenstate of

$$\begin{cases} L^2\,\psi_{n\ell m} = \hbar^2\,\ell(\ell+1)\,\psi_{n\ell m} \\ L_z\,\psi_{n\ell m} = \hbar m\,\psi_{n\ell m} \end{cases}$$

$$\Rightarrow \left(L_x^2 + L_y^2\right)\psi_{n\ell m} = \left[\hbar^2\,\ell(\ell+1) - (\hbar m)^2\right]\psi_{n\ell m}$$

$$= \hbar^2\left[\ell(\ell+1) - m^2\right]\psi_{n\ell m}$$

You'll get $\hbar^2\left(3(3+1) - 3^2\right) = \hbar^2\,(12-9) = 3\hbar^2$

with probability 1.

P 4.46

(a) $\ell = n-1$

then $$c_{j+1} = \frac{2\,(j + n-1+1-n)}{(j+1)\left(j + 2(n-1)+2\right)}\, c_j =$$

$$= \frac{2j}{(j+1)(j+2n)}\, c_j$$

Since $n = j_{max} + \ell + 1 = j_{max} + n-1+1 = j_{max} + n$

$\Rightarrow j_{max} = 0 \Rightarrow v(\rho) = C_0$ (pure constant)

Then, from (4.60), $u(\rho) = \rho^{\ell+1} e^{-\rho} v(\rho) = C_0 \rho^n e^{-\rho}$

But $\rho = \frac{r}{an}$ [eq. 4.73]

so that $$\frac{u(r)}{r} = R(r) = \frac{1}{r} C_0 \left(\frac{r}{an}\right)^n e^{-\frac{r}{an}} =$$

$$= \underbrace{\left(\frac{C_0}{(an)^n}\right)}_{N_n} r^{n-1} e^{-\frac{r}{an}}$$

Find normalization

$$1 = \int_0^\infty R^2 r^2 dr \Rightarrow 1 = N_n^2 \int_0^\infty \left(r^{n-1} e^{-\frac{r}{an}}\right)^2 r^2 dr$$

$$= N_n^2 \int_0^\infty r^{2n} e^{-\frac{2r}{an}} dr = N_n^2 (2n)! \left(\frac{an}{2}\right)^{2n+1}$$

Using $\int_0^\infty x^n e^{-x/a} dx = n!\, a^{n+1}$ from "Math. formulas" back page of the textbook
$= a^{n+1}\Gamma(n+1)$

$$\Rightarrow N_n = \left[\left(\frac{2}{an}\right)^{2n+1} \frac{1}{(2n)!}\right]^{1/2}$$

(b) $\psi_{n(n-1)m} = R_{n(n-1)} \, Y_{n-1}^{m}$

Calculating $\langle r \rangle$ or $\langle r^2 \rangle$ involves only radial functions as $\int |Y_\ell^m|^2 \sin\theta \, d\theta \, d\varphi = 1$ for any ℓ, m by construction.

Thus $\langle r \rangle = \int_0^\infty r \, N_n^2 \, r^{2(n-1)} e^{-\frac{2r}{na}} r^2 dr =$

$$= N_n^2 \int_0^\infty r^{2n+1} e^{-\frac{2r}{na}} dr = N_n^2 (2n+1)! \left(\frac{na}{2}\right)^{2n+2} =$$

$$= \frac{1}{(2n)!} \left(\frac{2}{an}\right)^{2n+1} \cdot (2n+1)! \left(\frac{na}{2}\right)^{2n+2} =$$

$$= (2n+1) \frac{na}{2} = n\left(n+\frac{1}{2}\right) a$$

Next, $\langle r^2 \rangle = \int r^2 \, N_n^2 \, r^{2(n-1)} e^{-\frac{2r}{na}} r^2 dr =$

$$= N_n^2 \int_0^\infty r^{2n+2} e^{-\frac{2r}{na}} dr = \frac{1}{(2n)!} \left(\frac{2}{an}\right)^{2n+1} \cdot$$

$$\cdot (2n+2)! \left(\frac{na}{2}\right)^{2n+3} = (2n+1)(2n+2) \left(\frac{na}{2}\right)^2 =$$

$$= \frac{1}{4} n^2 (2n+1)(2n+2) \, a^2 = n^2\left(n+\frac{1}{2}\right)(n+1) \, a^2 .$$

(c) $\sigma_r^2 = \langle r^2 \rangle - \langle r \rangle^2 = n^2\left(n+\frac{1}{2}\right)(n+1) \, a^2 - \left(n\left(n+\frac{1}{2}\right)a\right)^2 =$

$$= n^2 a^2 \left(n+\frac{1}{2}\right)\left((n+1) - \left(n+\frac{1}{2}\right)\right) = \frac{1}{2} n^2 \left(n+\frac{1}{2}\right) a^2$$

$$\Rightarrow \sigma_r = \frac{n\sqrt{\left(n+\frac{1}{2}\right)} \, a}{\sqrt{2}} = \frac{n\left(n+\frac{1}{2}\right)a}{\sqrt{2}\sqrt{n+\frac{1}{2}}} = \frac{\langle r \rangle}{\sqrt{2n+1}} \quad \text{indeed.}$$

P 4.57 let

$$q_1 = \frac{1}{\sqrt{2}}\left(x + \frac{a^2}{\hbar} p_y\right) \qquad p_1 = \frac{1}{\sqrt{2}}\left(p_x - \frac{\hbar}{a^2} y\right)$$

$$q_2 = \frac{1}{\sqrt{2}}\left(x - \frac{a^2}{\hbar} p_y\right) \qquad p_2 = \frac{1}{\sqrt{2}}\left(p_x + \frac{\hbar}{a^2} y\right)$$

(a) $[q_1, q_2] = 0$ because $[x, x] = [p_y, p_y] = [x, p_y] = 0$

the same for $[p_1, p_2] = 0$

But $[q_1, p_1] = \left[\frac{1}{\sqrt{2}}\left(x + \frac{a^2}{\hbar} p_y\right), \frac{1}{\sqrt{2}}\left(p_x - \frac{\hbar}{a^2} y\right)\right] =$

$= \frac{1}{2}\left([x, p_x] - [p_y, y]\right) = i\hbar.$

Similarly $[q_2, p_2] = i\hbar.$

Also useful to check that

$[q_1, p_2] = \frac{1}{2}\left[x + \frac{a^2}{\hbar} p_y, p_x + \frac{\hbar}{a^2} y\right] =$

$= \frac{1}{2}\left([x, p_x] + [p_y, y]\right) = \frac{1}{2}\left(i\hbar + (-i\hbar)\right) = 0.$

And $[q_2, p_1] = \frac{1}{2}\left[x - \frac{a^2}{\hbar} p_y, p_x - \frac{\hbar}{a^2} y\right] = \frac{1}{2}\left([x, p_x] + [p_y, y]\right) = 0$

Thus q_i and p_i are canonical position and momentum operators.

(b) $L_z = x p_y - y p_x$

from (a) $x = \frac{1}{\sqrt{2}}(q_1 + q_2) \qquad p_y = \frac{1}{\sqrt{2}}(q_1 - q_2)\frac{\hbar}{a^2}$

$y = \frac{1}{\sqrt{2}}(p_2 - p_1)\frac{a^2}{\hbar} \qquad p_x = \frac{1}{\sqrt{2}}(p_1 + p_2)$

Then $L_z = \frac{\hbar}{2a^2}(q_1 + q_2)(q_1 - q_2) - \frac{a^2}{2\hbar}(p_2 - p_1)(p_1 + p_2) = \frac{\hbar}{2a^2}(q_1^2 - q_2^2) + \frac{a^2}{2\hbar}(p_1^2 - p_2^2).$

(c) Thus $L_z = \frac{\hbar}{2a^2} q_1^2 + \frac{a^2}{2\hbar} p_1^2 - \left(\frac{\hbar}{2a^2} q_2^2 + \frac{a^2}{2\hbar} p_2^2\right) = H_1 - H_2$

Compare $H_{1,2}$ with the standard harmonic oscillator

$$H_{osc} = \frac{1}{2m} p^2 + \frac{1}{2} m \omega^2 x^2$$

$$H_1 = \frac{1}{2(\hbar/a^2)} p_1^2 + \frac{1}{2} \frac{\hbar}{a^2} q_1^2 \quad , \quad H_2 = \frac{1}{2(\hbar/a^2)} p_2^2 + \frac{1}{2} \frac{\hbar}{a^2} q_2^2$$

that is, both H_1 and H_2 are oscillators of "mass" $= \frac{\hbar}{a^2}$ and "frequency" $= 1$

(d) The eigenvalues of $H_{1,2}$ are $E_{1,2} = \hbar\left(n_{1,2} + \frac{1}{2}\right)$

$E_1 = \hbar\left(n_1 + \frac{1}{2}\right)$, $E_2 = \hbar\left(n_2 + \frac{1}{2}\right)$

$$\Rightarrow \quad L_z \psi = H_1 \psi - H_2 \psi = E_1 \psi - E_2 \psi = \hbar\left(n_1 + \tfrac{1}{2} - n_2 - \tfrac{1}{2}\right)\psi$$
$$= \hbar(n_1 - n_2)\psi$$

thus eigenvalues of L_z are $\hbar(n_1 - n_2) = \hbar \cdot (\text{integer})$

This makes eigenvalues of $L_z = \hbar m$ ↳ Integer